THERE IS NO k-INDEPENDENT ORDER OF MAGNITUDE FOR $r_k(N)$: A DEFECT IN A FORMAL STATEMENT OF ERDŐS PROBLEM #142

MATH0

ABSTRACT. Let $r_k(N)$ be the largest size of a subset of $\{1,\dots,N\}$ that contains no non-trivial k-term arithmetic progression. Erdős Problem #142 asks, for each fixed k, for an asymptotic formula for $r_k(N)$, and in a weaker form for its order of magnitude; the formalization `Erdos142.erdos_142.variants.isTheta` of the order-of-magnitude question in the formal-conjectures repository instead asks for a single function $f\colon \mathbb{N}\to\mathbb{R}$, not depending on k, with $r_k \asymp f$ for every $k\in\mathbb{N}$. We prove that no such function exists, because $r_1(N)=N$, $r_2(N)\le 1$, and $\asymp$ is an equivalence relation. This is a misformalization report, not progress on the research problem: the argument uses only the degenerate cases $k=1,2$, the problem remains open for every $k\ge 3$, and we propose a corrected statement in which the unknown depends on k. The refutation, valid for every value of the unknown, is verified in Lean 4 against the upstream formal-conjectures statement at commit `c77fdd4`.

1. Introduction

For $k\ge 1$ and $N\ge 0$ let $r_k(N)$ be the largest size of a subset of $\{1,\dots,N\}$ containing no non-trivial arithmetic progression of length k. Erdős Problem #142 [3] asks for an asymptotic formula for $r_k(N)$; the problem page records that Erdős also asked for the order of magnitude of $r_k(N)$ [4], lists a prize of \$10000, and states that an asymptotic formula is out of reach even for $k=3$. The first values of $r_3,\dots,r_6$ are the OEIS sequences A003002–A003005 [17].

Known results. Szemerédi proved that $r_k(N)=o(N)$ for every fixed k [20], and Gowers gave the first reasonable quantitative bounds for general k, $r_k(N)\le N(\log\log N)^{-c_k}$ [10]. Much stronger upper bounds are now known: $r_3(N)\le N\exp(-c(\log N)^{1/12})$ by Kelley and Meka [12], with the exponent improved to $1/9$ by Bloom and Sisask [2] (a 2026 preprint of Raghavan [19] announces $r_3(N)\le N\exp(-c(\log N)^{1/6}(\log\log N)^{-1})$); $r_4(N)\le N(\log N)^{-c}$ by Green and Tao [11]; and $r_k(N)\le N\exp(-(\log\log N)^{c_k})$ for $k\ge 5$ by Leng, Sah and Sawhney [13]. In the other direction, Behrend's construction gives $r_3(N)\ge N\exp(-c\sqrt{\log N})$ [1], and Rankin generalized the construction to all k [18]; in the explicit form given by O'Bryant [16], for $k\ge 3$ and $n=\lceil\log_2 k\rceil$ one has $r_k(N)\ge N\exp(-c_k(\log N)^{1/n})$ for large N. The order of magnitude of $r_k(N)$ is unknown for every $k\ge 3$.

The formal statement. The formal-conjectures repository [5] formalizes Problem #142 in `FormalConjectures/ErdosProblems/142.lean`. The docstring of `erdos_142.variants.isTheta` reads "find f_k with $r_k(N)\asymp_k f_k(N)$", so the intended unknown is a family of functions indexed by k. In the formal statement, however, the unknown is a top-level definition `erdos_142.variants.isTheta.answer` of type $\mathbb{N}\to\mathbb{R}$, which cannot depend on k, while the theorem quantifies over all $k\in\mathbb{N}$, including $k=0,1,2$. The quantifier pattern $\forall k\,\exists f_k$ of the problem has become $\exists f\,\forall k$.

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Contribution and status. We prove that the formal statement is false whatever value is chosen for its unknown (Theorem 1) and verify this in Lean 4. The proof uses only $r_1(N) = N$ and $r_2(N) \le 1$, so it says nothing about r_k for $k \ge 3$; we classify it as a *misformalization report*.

The difficulty has been noticed upstream but not fixed. When the hypothesis $1 < k$ was added to `erdos_142.variants.lower`, the description of pull request #1187 [6] noted that $r_0(N) = r_1(N) = N$ and left the statements with an `answer(sorry)` slot unchanged. Pull request #5446 [7], opened on 10 September 2026 and still open, concerns the mode in which an answer is extracted as a closed definition, where "a slot typed without that binder can never be filled in correctly"; it lists lines 37 and 56 of `142.lean` among further slots with this defect that it does not fix. At that time these lines held `erdos_142`, then stated with `=Θ[atTop]` and hence of the same shape as the statement refuted here, and `erdos_142.variants.upper`, stated with `=O[atTop]`. Two days later pull request #5810 [8] changed `erdos_142` to `~[atTop]` and replaced `erdos_142.variants.upper` by the present Θ-variant, keeping the type $\mathbb{N} \to \mathbb{R}$ of the unknown; pull request #7327 [9] (14 October 2026) turned the `answer(sorry)` slot into the closed definition `erdos_142.variants.isTheta.answer`. Before that change, under the default elaboration an `answer(sorry)` slot could be filled with a term mentioning the binder k (the description of #5810 notes that the tautological answer `r k` closes these statements); since #7327 the unknown is a closed constant, and this is the statement refuted here. We found no earlier formal refutation of the statement. The same argument refutes `erdos_142`, whose unknown is also independent of k; math0 records that refutation separately.

2. Statement

2.1. **Definitions.** Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. Following the library of [5], a set $P \subseteq \mathbb{N}$ is an *arithmetic progression of length* $l \in \mathbb{N}$ if there are $a, d \in \mathbb{N}$ with $P = \{a + nd : 0 \le n < l\}$ and $|P| = l$. For $l \ge 2$ the cardinality condition forces $d \ge 1$, so these are exactly the non-trivial progressions. A set $A \subseteq \mathbb{N}$ is *free of progressions of length* l if every arithmetic progression of length l contained in A satisfies $l \le 1$. For $l \ge 2$ this says that A contains no arithmetic progression of length l; for $l \in \{0, 1\}$ the condition is vacuous. For $k, N \in \mathbb{N}$ put

$$r_k(N) = \sup\bigl\{|S| : S \subseteq \{1, \dots, N\},\ S \text{ free of progressions of length } k\bigr\},$$

the supremum being taken in $\mathbb{N}$; the set is non-empty (it contains $|\emptyset| = 0$) and bounded by N. For $k \ge 3$ this is the quantity of Problem #142.

For $f, g \colon \mathbb{N} \to \mathbb{R}$ we write $f = O(g)$ if there are $C \in \mathbb{R}$ and $N_0 \in \mathbb{N}$ with $|f(N)| \le C|g(N)|$ for all $N \ge N_0$, and $f \asymp g$ if $f = O(g)$ and $g = O(f)$. These are Mathlib's `Asymptotics.IsBigO` and `Asymptotics.IsTheta` along `Filter.atTop`, written `=O[atTop]` and `=Θ[atTop]`. Since O is reflexive and transitive, $\asymp$ is an equivalence relation.

2.2. **The upstream statement.** In `FormalConjecturesForMathlib/Combinatorics/AP/Basic.lean` at commit `c77fdd4` the definitions read as follows (docstrings omitted; α is a commutative additive monoid and $\mathbb{N}\infty$ is $\mathbb{N} \cup \{\infty\}$):

```
def Set.IsAPOfLengthWith (s : Set α) (l : ℕ∞) (a d : α) : Prop :=
  ENat.card s = l ∧ s = {a + n • d | (n : ℕ) (_ : n < l)}

def Set.IsAPOfLength (s : Set α) (l : ℕ∞) : Prop :=
  ∃ a d : α, s.IsAPOfLengthWith l a d

def Set.IsAPOfLengthFree (s : Set α) (l : ℕ∞) : Prop :=
  ∀ t ⊆ s, t.IsAPOfLength l → l ≤ 1

noncomputable def Set.IsAPOfLengthFree.maxCard (k : ℕ) (N : ℕ) : ℕ :=
```

```
  sSup {Finset.card S | (S) (_ : S ⊆ Finset.Icc 1 N) (_ : (S : Set ℕ).
    IsAPOfLengthFree k)}
```

The relevant declarations, copied verbatim from `FormalConjectures/ErdosProblems/142.lean` (inside `namespace Erdos142`, with `open Asymptotics Filter`; the other declarations of the file are omitted), are:

```
noncomputable abbrev r := Set.IsAPOfLengthFree.maxCard

def erdos_142.variants.isTheta.answer : ℕ → ℝ := sorry

/--
Determine the order of magnitude of $r_k(N)$, the largest possible size of a subset
of $\{1, \dots, N\}$ that does not contain any non-trivial $k$-term arithmetic
    progression.
That is, find $f_k$ with $r_k(N) \asymp_k f_k(N)$.
-/
@[category research open, question value, AMS 11]
theorem erdos_142.variants.isTheta (k : ℕ) :
    (fun N => (r k N : ℝ)) =Θ[atTop] erdos_142.variants.isTheta.answer := by
  sorry
```

Writing f for the value of the unknown, the statement asserts

$$P(f): \qquad r_k \asymp f \quad \text{for every } k \in \mathbb{N}, \tag{1}$$

where r_k is the function $N \mapsto r_k(N)$, viewed as real-valued.

2.3. **Main result.**

Theorem 1. *For every $f\colon \mathbb{N} \to \mathbb{R}$ the statement $P(f)$ is false. More precisely, there is no $f\colon \mathbb{N} \to \mathbb{R}$ with both $r_1 \asymp f$ and $r_2 \asymp f$.*

In Lean, Theorem 1 is the following declaration (in `namespace Math0.Erdos142`); its type is the negation of the upstream statement, universally quantified over the unknown.

```
theorem not_isTheta : ∀ a : ℕ → ℝ,
    ¬ ∀ k : ℕ, (fun N => (Erdos142.r k N : ℝ)) =Θ[atTop] a
```

3. Proof

Lemma 2. *$r_1(N) = N$ for every $N \in \mathbb{N}$.*

Proof. Every set is free of progressions of length 1, so the set in the definition of $r_1(N)$ is $\{|S| : S \subseteq \{1, \dots, N\}\} = \{0, 1, \dots, N\}$, whose supremum is N, attained by $S = \{1, \dots, N\}$. □

In Lean this is the library lemma `Set.IsAPOfLengthFree.maxCard_one` of [5].

Lemma 3. *Let $S \subseteq \mathbb{N}$ be a finite set that is free of progressions of length 2. Then $|S| \le 1$.*

Proof. Suppose S has two distinct elements, and call them $a < b$. With $d = b - a \ge 1$ we have $\{a, b\} = \{a + nd : 0 \le n < 2\}$ and $|\{a, b\}| = 2$, so $\{a, b\}$ is an arithmetic progression of length 2 contained in S. Freeness gives $2 \le 1$, a contradiction. Hence any two elements of S are equal, that is, $|S| \le 1$. □

The Lean proof (`isTheta_card_le_one_of_free`) rewrites the goal with `Finset.card_le_one`, splits $a \neq b$ into the cases $a < b$ and $b < a$, and in each case applies the freeness hypothesis to the pair, using the library lemma `Nat.isAPOfLength_pair`: if $a < b$ then $\{a, b\}$ is an arithmetic progression of length 2.

Corollary 4. *$r_2(N) \leq 1$ for every $N \in \mathbb{N}$.*

Proof. By Lemma 3, 1 is an upper bound for the set in the definition of $r_2(N)$, so its supremum in $\mathbb{N}$ is at most 1. □

In Lean (`isTheta_r_two_le_one`) the last step is `csSup_le'`, which bounds the supremum of a set of natural numbers by any upper bound of the set. (In fact $r_2(N) = \min(N, 1)$, but only the upper bound is needed.)

Proof of Theorem 1. Let $f\colon \mathbb{N} \to \mathbb{R}$ and suppose that $r_1 \asymp f$ and $r_2 \asymp f$; this holds in particular if $P(f)$ holds. By symmetry and transitivity of $\asymp$ we get $r_1 \asymp r_2$, and in particular $r_1 = O(r_2)$. By Corollary 4, $|r_2(N)| \leq 1 \cdot |1|$ for all N, so $r_2 = O(1)$, where 1 is the constant function. By transitivity of O, $r_1 = O(1)$, which by Lemma 2 says that $N \mapsto N$ is $O(1)$: there are $C \in \mathbb{R}$ and $N_0 \in \mathbb{N}$ with $N \leq C$ for all $N \geq N_0$. This fails for any integer $N \geq \max(N_0, C + 1)$, a contradiction. □

The Lean proof of `not_isTheta` (reproduced in Section 5) follows these steps. It obtains $r_1 = O(r_2)$ as `((h 1).trans (h 2).symm).isBigO`, proves $r_2 = O(1)$ with `IsBigO.of_bound` and the constant 1, composes the two, and rewrites r_1 with `maxCard_one`. The lemma `isBigO_one_iff` turns $N = O(1)$ into the statement that $N \mapsto \|N\|$ is eventually bounded above, and `Filter.not_isBoundedUnder_of_tendsto_atTop` refutes this, since $\|N\| \to \infty$ by `tendsto_norm_atTop_atTop` composed with `tendsto_natCast_atTop_atTop`.

4. The defect and a corrected statement

By Theorem 1, `erdos_142.variants.isTheta` cannot be proved for any value of its unknown, and its negation is provable for every value. Its falsity comes from the degenerate cases $k \leq 2$ together with the fact that the unknown cannot depend on k: every set is free of progressions of length 0 and 1, and no set with two elements is free of progressions of length 2. The intended problem, to find for each $k \geq 3$ an explicit f_k with $r_k \asymp f_k$, is not touched by this. Restricting the formal statement to $k \geq 3$ while keeping a k-independent unknown would not make it satisfiable either.

Remark 5. Let $k \geq 513$, so that $n = \lceil \log_2 k \rceil \geq 10$. Rankin's bound in the form of [16] gives $r_k(N) \geq N \exp(-c_k(\log N)^{1/10})$ for large N, while $r_3(N) \leq N \exp(-c(\log N)^{1/9})$ [2]. Hence $r_3(N)/r_k(N) \leq \exp\big(c_k(\log N)^{1/10} - c(\log N)^{1/9}\big) \to 0$, so $r_3 \not\asymp r_k$, and no single f satisfies $r_k \asymp f$ for all $k \geq 3$. This uses the cited bounds and is not part of the formal verification. Whether $r_k(N)/r_{k+1}(N) \to 0$ for some $k \geq 3$ is a different question, which Erdős noted as open in [4] (see [3]).

We propose the following corrected statement, in which the unknown is indexed by k and the degenerate cases are excluded:

```
def erdos_142.variants.isTheta.answer : ℕ → ℕ → ℝ := sorry

theorem erdos_142.variants.isTheta (k : ℕ) (hk : 3 ≤ k) :
    (fun N => (r k N : ℝ)) =Θ[atTop] erdos_142.variants.isTheta.answer k := by
  sorry
```

We checked that this elaborates against the pinned Lean, Mathlib and formal-conjectures versions of Section 5. Once the unknown depends on k, the hypothesis $3 \le k$ is no longer needed for satisfiability (for $k \le 2$ the functions r_k are known explicitly), but it matches the intended problem. The same change, giving `erdos_142.answer` the type $\mathbb{N} \to \mathbb{N} \to \mathbb{R}$ and applying it to k, repairs the main statement `erdos_142`. As the description of pull request #5810 [8] notes for value questions of this shape, the tautological value `answer` $k = r_k$ satisfies such a statement; the repository's contribution guidelines treat this as out of scope, and it is not specific to Problem #142.

5. Machine verification

Files and declarations. The proof is the file `lean/Math0/ErdosProblems/Erdos142_erdos_142_variants_isTheta.lean` of the math0 repository (`https://github.com/Foundation-0/math0`). It imports only the upstream file `FormalConjectures/ErdosProblems/142.lean` and the audit commands `lean/Math0/Audit.lean`. Its declarations, in the namespace `Math0.Erdos142`, are `isTheta_card_le_one_of_free` (Lemma 3), `isTheta_r_two_le_one` (Corollary 4) and the main theorem `not_isTheta` (Theorem 1). The file contains no `sorry`, no `native_decide` and no new axioms. The main proof and the final audit command are:

```
theorem not_isTheta : ∀ a : ℕ → ℝ,
    ¬ ∀ k : ℕ, (fun N => (Erdos142.r k N : ℝ)) =Θ[atTop] a := by
  intro a h
  have h12 : (fun N => (Erdos142.r 1 N : ℝ)) =O[atTop] (fun N => (Erdos142.r 2 N : ℝ)
    ) :=
    ((h 1).trans (h 2).symm).isBigO
  have h2b : (fun N => (Erdos142.r 2 N : ℝ)) =O[atTop] (fun _ => (1 : ℝ)) := by
    apply Asymptotics.IsBigO.of_bound 1
    filter_upwards with N
    simp only [Real.norm_eq_abs, abs_one, mul_one, Nat.abs_cast]
    exact_mod_cast isTheta_r_two_le_one N
  have h3 := h12.trans h2b
  simp only [Erdos142.r, Set.IsAPOfLengthFree.maxCard_one] at h3
  rw [Asymptotics.isBigO_one_iff] at h3
  exact Filter.not_isBoundedUnder_of_tendsto_atTop
    (tendsto_norm_atTop_atTop.comp tendsto_natCast_atTop_atTop) h3

#math0_refutes_every_answer Math0.Erdos142.not_isTheta Erdos142.erdos_142.variants.
    isTheta with Erdos142.erdos_142.variants.isTheta.answer
```

Audit. The command `#math0_refutes_every_answer`, defined in `lean/Math0/Audit.lean`, is the math0 audit for misformalization reports on value questions. It fails unless: (1) our theorem has the same universe parameters as the upstream theorem, and the unknown is not universe-polymorphic; (2) the type of our theorem is syntactically identical, up to metadata annotations and with no unfolding or definitional-equality check, to $\forall a,\ \neg B[a]$, where $B[a]$ is the type of the upstream theorem with every occurrence of the constant `Erdos142.erdos_142.variants.isTheta.answer` replaced by a bound variable a of the same type $\mathbb{N} \to \mathbb{R}$; (3) our theorem depends on no axioms other than `propext`, `Classical.choice` and `Quot.sound`, so `sorryAx`, the axioms behind `native_decide` and custom axioms are rejected. Condition (2) ensures that the theorem refutes the upstream statement as published, for every value of its unknown. On this file the command reports

```
math0 audit: 'Math0.Erdos142.not_isTheta' depends on axioms [Classical.choice, Quot.
    sound, propext]
math0 statement check: 'Math0.Erdos142.not_isTheta' refutes 'Erdos142.erdos_142.
    variants.isTheta' for every value of 'Erdos142.erdos_142.variants.isTheta.answer'
```

Toolchain and reproduction. We use Lean 4 [15] `v4.33.1`, Mathlib [14] `v4.33.1` and formal-conjectures [5] at commit `c77fdd446d23539302aa73bcfae4a9fa962b8382`, all pinned in `lean/`. To reproduce, run `scripts/verify.sh` from the repository root: it fetches the Mathlib build cache, builds all math0 results with `lake build` (re-running every audit command), checks that the `formal_conjectures` and `mathlib` checkouts are unmodified, and reports the number of passed statement checks. The single file can also be checked with `lake env lean` from the `lean` directory.

References

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[6] formal-conjectures, pull request #1187 (commit `bf0ea56`), *fix: exclude 0 and 1 cases when using* `Set.IsAPOfLengthFree.maxCard`, 11 November 2025, `https://github.com/google-deepmind/formal-conjectures/pull/1187`.

[7] formal-conjectures, pull request #5446, *fix: answer slots that cannot depend on the inputs they must depend on*, opened 10 September 2026, `https://github.com/google-deepmind/formal-conjectures/pull/5446`.

[8] formal-conjectures, pull request #5810 (commit `7f64240`), *fix(Erdos/142): ask for asymptotic formula; order of magnitude variant*, 12 September 2026, `https://github.com/google-deepmind/formal-conjectures/pull/5810`.

[9] formal-conjectures, pull request #7327 (commit `2c24ebe`), *feat(Util) replace* `answer(sorry)` *with* `def answer`, 14 October 2026, `https://github.com/google-deepmind/formal-conjectures/pull/7327`.

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Produced by Claude (an AI model developed by Anthropic) in the math0 project